

Linux SSH Brute Force Detection and Triage Using Splunk

This project demonstrates the creation of a SOC-style monitoring workflow using Splunk Enterprise and Ubuntu Linux. Failed SSH authentication attempts were intentionally generated from a Windows system and forwarded into Splunk for detection, alerting, and triage analysis.

Skills Demonstrated
Splunk Enterprise
Linux Authentication Log Analysis
Security Monitoring and Alerting
SOC Alert Triage
SSH Brute Force Detection
Basic SPL Query Development

Lab Environment

- Ubuntu Linux host configured with SSH access
- Splunk Enterprise running in a Docker container
- Splunk ingesting authentication logs from /var/log/auth.log
- Windows workstation used to simulate failed SSH authentication attempts

Attack Simulation

Failed SSH login attempts were intentionally generated from a Windows system using PowerShell. Incorrect passwords were entered multiple times to simulate brute-force authentication behavior.

```
Windows PowerShell
jake@192.168.1.84
jake@192.168.1.84:~$ ssh -p 2222 jake@192.168.1.84
Permission denied, please try again.
jake@192.168.1.84's password:
jake@192.168.1.84: Permission denied (publickey,password).
jake@192.168.1.84:~$ ssh -p 2222 jake@192.168.1.84
jake@192.168.1.84's password:
Permission denied, please try again.
jake@192.168.1.84's password:
Permission denied, please try again.
jake@192.168.1.84's password:
jake@192.168.1.84: Permission denied (publickey,password).
```

Detection Logic

A Splunk SPL query was developed to identify excessive failed SSH authentication attempts. The detection logic counted matching failed login events and triggered an alert once the threshold exceeded five events.

splunk> Enterprise

Search & Reporting

Search

Analytics

Datasets

Reports

Alerts

Dashboards

Modules

Linux SSH Brute Force Detection

SPL

Convert to SPL2

Save

Save As

View

Create Table View

Close

index=main source="/var/log/auth.log" "failed password"

| stats count

| where count >= 5

6 of 12 events matched

No Event Sampling

Job

II

Verbose Mode

Events (6)

Patterns

Statistics (1)

Visualization

Show: 20 Per Page

Format

count

6

Authentication Log Analysis

Splunk successfully ingested and displayed failed SSH login events from the Ubuntu authentication log. Event analysis included timestamps, usernames, and originating IP addresses.

splunk> Enterprise

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Time range: Real-time

6 of 12 events matched

No Event Sampling

Job

Verbose Mode

Events (6)

Patterns

Statistics

Visualization

Timeline format

Zoom Out

Zoom to Selection

Deselect

Format

Show: 20 Per Page

View List

Hide Fields

All Fields

SELECTED FIELDS

host 1

source 1

sourcetype 1

INTERESTING FIELDS

date_hour 1

date_mday 1

date_minute 1

date_month 1

date_second 6

date_wday 1

date_year 1

date_zone 1

index 1

linecount 1

punct 1

splunk_server 1

timeendpos 1

timestartpos 1

Extract New Fields

i	Time	Event
>	5/22/26 11:51:53.113 PM	2026-05-22T19:51:53.113136-04:00 VivoBook-ASUSLaptop sshd-session[223269]: Failed password for jake from 192.168.1.138 port 50309 ssh2 host = VivoBook-ASUSLaptop source = /var/log/auth.log sourcetype = auth
>	5/22/26 11:51:48.302 PM	2026-05-22T19:51:48.302116-04:00 VivoBook-ASUSLaptop sshd-session[223269]: Failed password for jake from 192.168.1.138 port 50309 ssh2 host = VivoBook-ASUSLaptop source = /var/log/auth.log sourcetype = auth
>	5/22/26 11:51:43.691 PM	2026-05-22T19:51:43.691242-04:00 VivoBook-ASUSLaptop sshd-session[223269]: Failed password for jake from 192.168.1.138 port 50309 ssh2 host = VivoBook-ASUSLaptop source = /var/log/auth.log sourcetype = auth
>	5/22/26 11:51:37.303 PM	2026-05-22T19:51:37.303643-04:00 VivoBook-ASUSLaptop sshd-session[222894]: Failed password for jake from 192.168.1.138 port 64016 ssh2 host = VivoBook-ASUSLaptop source = /var/log/auth.log sourcetype = auth
>	5/22/26 11:51:31.306 PM	2026-05-22T19:51:31.306784-04:00 VivoBook-ASUSLaptop sshd-session[222894]: Failed password for jake from 192.168.1.138 port 64016 ssh2 host = VivoBook-ASUSLaptop source = /var/log/auth.log sourcetype = auth
>	5/22/26 11:51:27.249 PM	2026-05-22T19:51:27.249826-04:00 VivoBook-ASUSLaptop sshd-session[222894]: Failed password for jake from 192.168.1.138 port 64016 ssh2 host = VivoBook-ASUSLaptop source = /var/log/auth.log sourcetype = auth

Alert Configuration

A real-time Splunk alert named **Linux SSH Brute Force Detection** was configured to monitor failed SSH authentication attempts occurring within a five-minute window.

Save As Alert

Title

Linux SSH Brute Force Detection

Description

Triggers when multiple failed SSH authentication attempts are detected within a short time window on the monitored Ubuntu host.

Permissions

Private

Shared in App

Alert type

Scheduled

Real-time

Expires

5

minute(s)

Trigger Conditions

Trigger alert when

Number of Results

is greater than

0

in

5

minute(s)

Trigger

Once

For each result

Throttle

Trigger Actions

+ Add Actions

When triggered

Severity

High

Remove

Cancel

Save

Triggered Alert Results

After generating repeated failed login attempts, the Splunk alert triggered successfully. The alert history confirmed multiple detections across the monitored timeframe.

splunk> Enterprise

Search & Reporting

Search

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Modules

Linux SSH Brute Force Detection

Triggers when multiple failed SSH authentication attempts are detected within a short time window on the monitored Ubuntu host.

Enabled: Yes, Disable

App: search

Permissions: Private. Owned by admin, Edit

Modified: May 22, 2026 11:49:21 PM

Alert Type: Real-time, Edit

Trigger Condition: Number of Results is > 0 in 5 minutes, Edit

Actions: 2 Actions

Add to Triggered Alerts

Add to Triggered Alerts

Trigger History

20 per page

	TriggerTime	Actions
1	2026-05-22 23:52:19 UTC	View Results
2	2026-05-22 23:52:14 UTC	View Results
3	2026-05-22 23:52:09 UTC	View Results
4	2026-05-22 23:52:04 UTC	View Results
5	2026-05-22 23:51:59 UTC	View Results
6	2026-05-22 23:51:54 UTC	View Results
7	2026-05-22 23:51:53 UTC	View Results
8	2026-05-22 23:51:48 UTC	View Results
9	2026-05-22 23:51:45 UTC	View Results
10	2026-05-22 23:51:44 UTC	View Results
11	2026-05-22 23:51:39 UTC	View Results
12	2026-05-22 23:51:38 UTC	View Results
13	2026-05-22 23:51:33 UTC	View Results
14	2026-05-22 23:51:32 UTC	View Results

Findings and Conclusion

Splunk successfully detected repeated failed SSH login attempts against the Ubuntu host. The project demonstrated a complete SOC-style workflow including attack simulation, log ingestion, alert creation, detection tuning, and alert triage.

This homelab strengthened practical understanding of:

- Security monitoring workflows
- Linux authentication logging
- Splunk search processing language (SPL)
- Alert configuration and threshold tuning
- SOC investigation and event triage